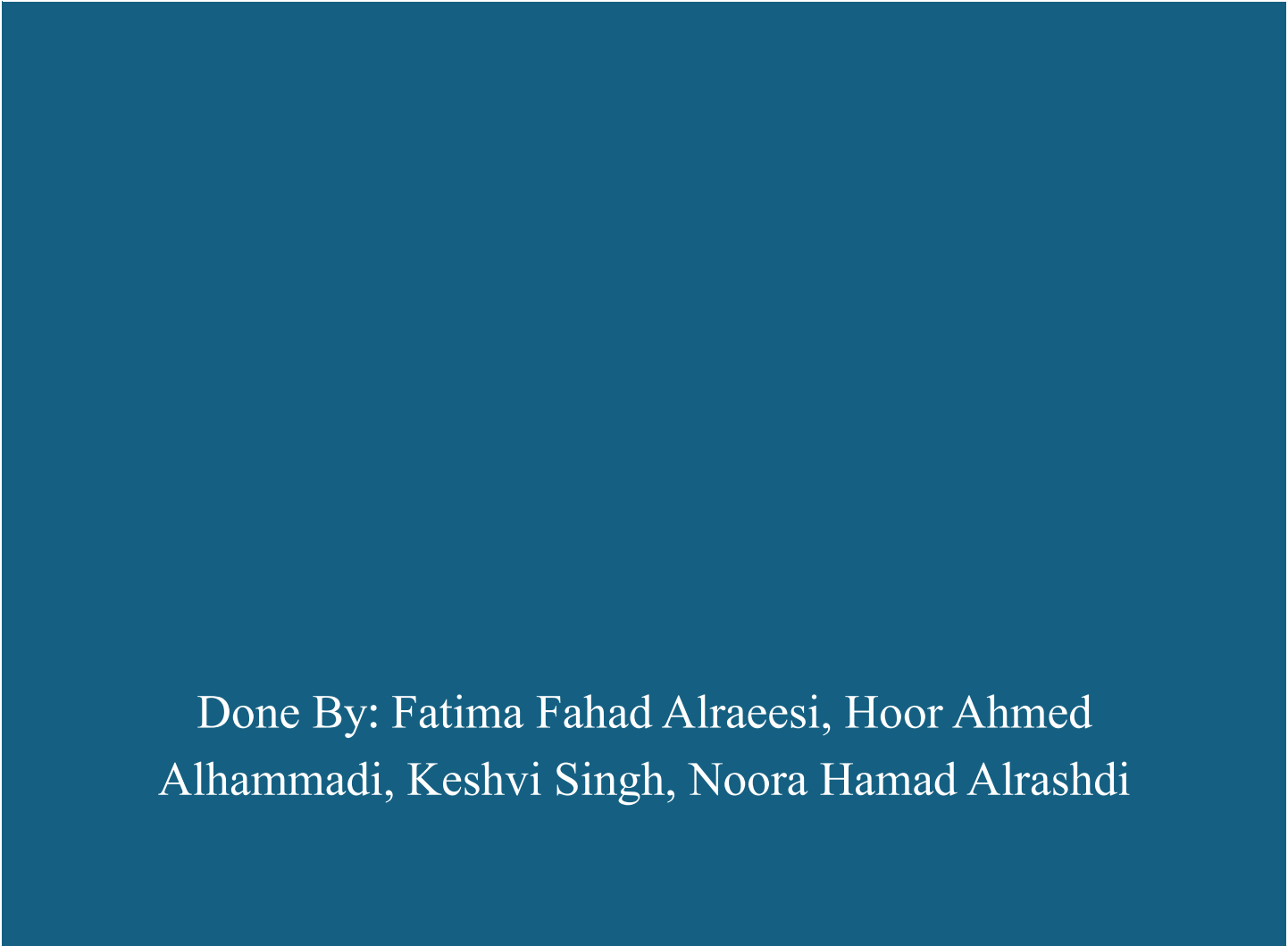




AELL 101 NOTES

LECTURES 1-3



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COMPONENTS OF ELECTRICAL CIRCUIT

BASIC QUANTITIES

1) Charge: -

- It is the most basic electrical quantity
- Property of atoms
- It is measured in coulombs (C)

$$Q = n \times e$$

Here, Q is electric charge, n is the number of electrons, and e is the charge of one electron ($1.6 \times 10^{-19} \text{ C}$)

The number of electrons in one coulomb can be calculated,

$$1 e = 1.6 \times 10^{-19} \text{ C}$$

$$\therefore 1 \text{ C} = \frac{1}{1.6 \times 10^{-19}} = 6.25 \times 10^{18} \text{ electrons}$$

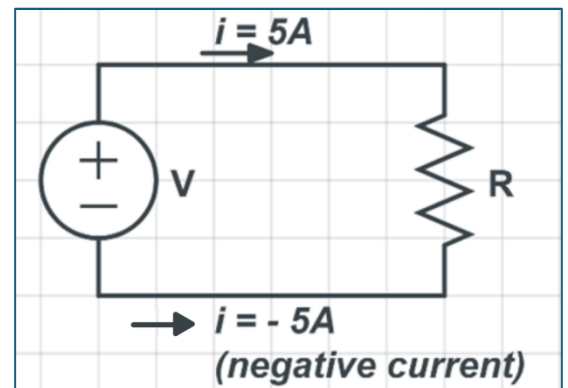
2) Current: -

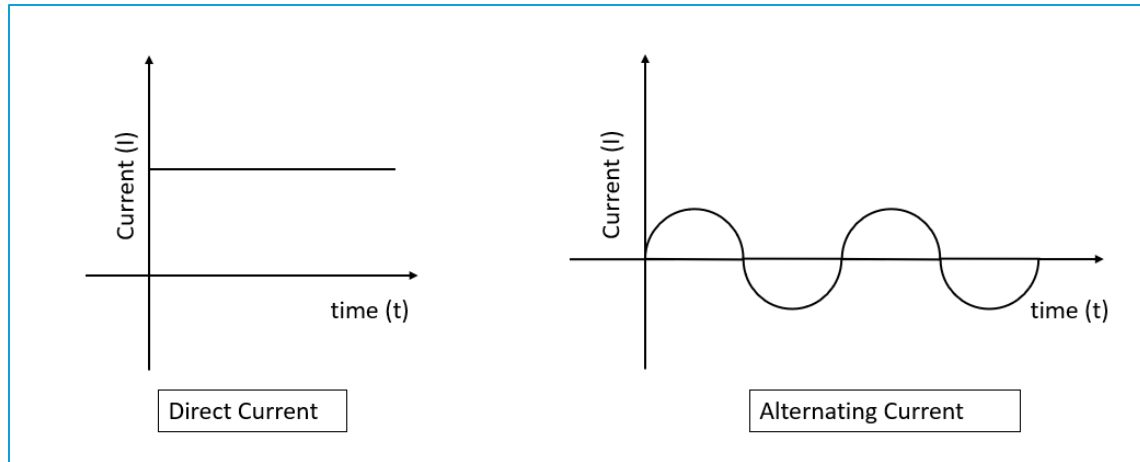
- It is the rate of change of charge
- The flow of current (positive to negative) is opposite to the flow of electrons (negative to positive)
- It is measured in Amperes (A)

$$1 \text{ A} = 1 \text{ C/s}$$

- There are two types of currents:

1. Alternating Current (AC)
2. Direct Current (DC)





3) Voltage: -

- It is the energy required to move a unit of charge from the reference point (-) to another point (+)
- It is measured in volts (V) or Joules per coulomb

$$V = \frac{dW}{dQ}$$

4) Power: -

- It is the rate of change of energy absorbed or expended
- Measured in Watts (W)

$$P = \frac{dW}{dt} = \frac{dW}{dQ} \times \frac{dQ}{dt}$$

$$\therefore P = V \times i$$

5) Energy: -

- It is the quantitative property that is transferred to a body or a physical system

- Equivalent to work done (W) on or by the system
- Measured in watt-hour or kilowatt-hour

$$W = \int_{t_0}^t P dt = \int_{t_0}^t (Vi) dt$$

Related Examples

Example 1:

If a home has 4 bulbs of 60 watts each, 2 fans of 20 watts each, and a refrigerator of 200 watts, and if the bulbs are turned on for 6 hours a day, the fans are turned on for 5 hours a day, and the refrigerator for 24 hours a day, what is the electricity bill for the month of January?

Solution:

Total energy from bulbs = $P \times t \times n = 60 \times 6 \times 4 = 1440 \text{ wh}$

Total energy from fan = $P \times t \times n = 20 \times 5 \times 2 = 200 \text{ wh}$

Total energy from refrigerator = $P \times t = 200 \times 24 = 4800 \text{ wh}$

T.E/day = $1440 + 200 + 4800 = 6440 \text{ wh} = 6.44 \text{ kwh}$

$\therefore \text{T.E for January} = 6.44 \times 31 = 199.64 \text{ kwh}$

$\therefore \text{Total cost} = 199.64 \text{ kwh} \times 0.2 \text{ AED/kwh} = 39.928 \text{ AED}$

Example 2:

Calculate the power supplied to a beam of 50 million billion electrons per second passing through a voltage difference of 20,000 volts.

Solution:

$$i = n \times e = 50 \times 10^6 \times 10^9 \times 1.6 \times 10^{-19} = 8 \times 10^{-3} A$$

$$P = V \times i = 20000 \times 8 \times 10^{-3} = 1.6 W$$

$$\therefore \text{Power supplied by beam} = 1.6 W$$

Example 3:

The total charge entering a terminal is given by

$$Q = 5t \sin(4\pi t) \text{ mC}.$$

Calculate the current at $t = 0.5$ seconds.

Solution:

$$i = \frac{dQ}{dt} = \frac{d(5t \sin(4\pi t))}{dt} = 5 \sin(4\pi t) + 5t \cos(4\pi t) 4\pi$$

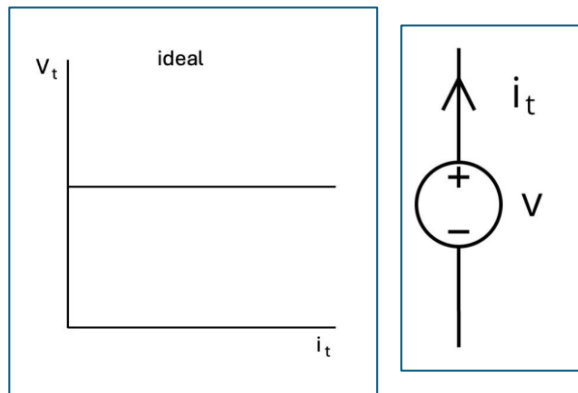
$$\therefore i(t = 0.5s) = 5 \sin(2\pi) + \frac{5}{2} \cos(2\pi) 4\pi = 31.42 \text{ mA}$$

BASIC CIRCUITAL ELEMENTS

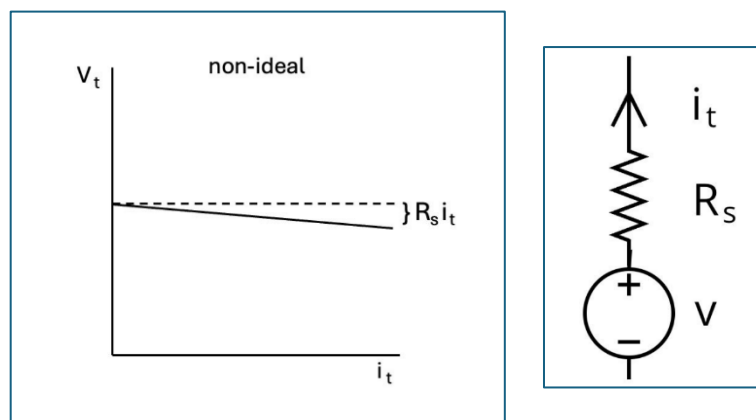
1) Sources: -

- Independent Voltage Source:

- An independent voltage source is able to supply and maintain the same voltage across its terminals regardless of the current flowing through it.
- **Ideal Voltage Source** – It is the type of voltage source that has a straight line I-V characteristic.

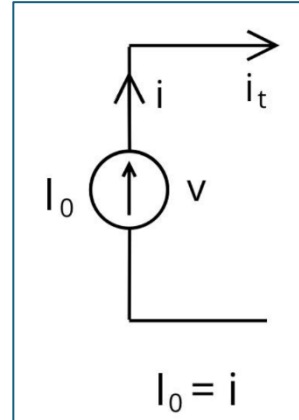
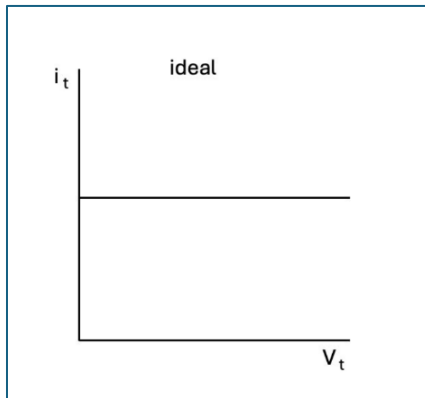


- **Non-Ideal Voltage Source** – Unlike ideal voltage source, non-ideal voltage source will have a non-linear I-V characteristic due to presence of internal source resistance (or impedance) R_s .

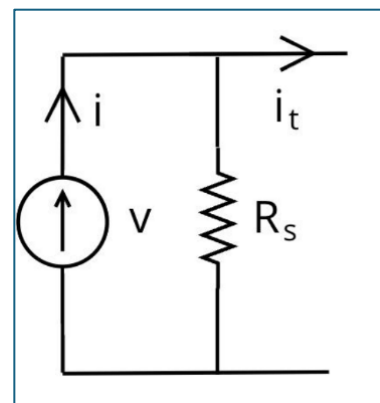
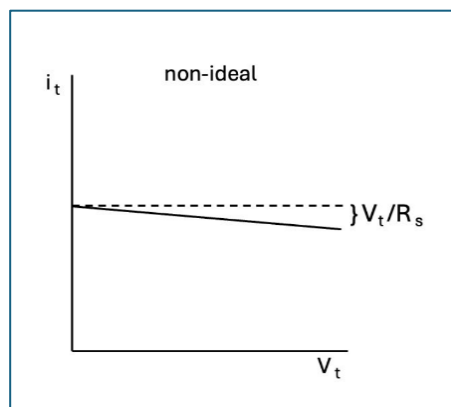


- **Independent Current Source:**

- It is a current source that provides steady current independent of the load connected to it.
- **Ideal Current Source** – It has a straight line I-V characteristic (i does not depend on load)



- **Non-Ideal Current Source** – The behavior of non-ideal current source is different from ideal source due to presence of internal source resistance (or impedance) R_s . This causes the I-V characteristic of source to be non-linear.



2) Resistance(R): -

- It is the ability of the material to resist the flow of current.
- It is measured in ohms (Ω)

$$V = i \times R$$

- **Conductance:**

- Inverse of resistance

- It is measured in moh ($\text{m}\Omega$), siemens (S), ohm^{-1}

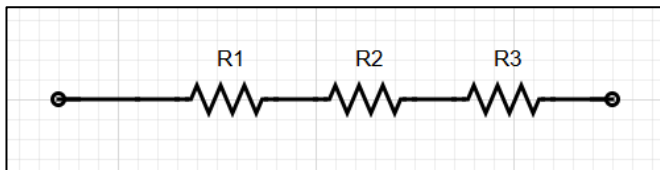
$$G = \frac{1}{R}$$

- **Power:**

- $P = V \times i = i^2 R = \frac{V^2}{R}$

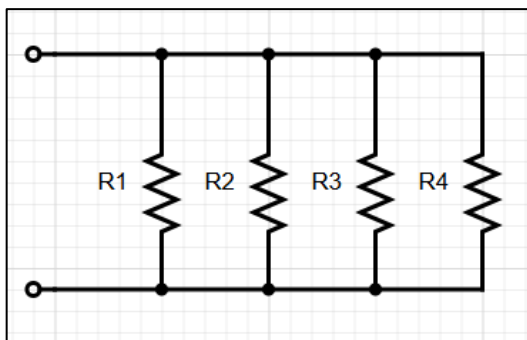
- **Series Connection:**

- $R_{eq} = R_1 + R_2 + \dots + R_n$



- **Parallel Connection:**

- $R_{eq} = 1 / (1/R_1 + 1/R_2 + \dots + 1/R_n)$



3) Capacitor: -

- It is a device that stores energy in the form of charge.
- Capacitance (C) is the ability of a capacitor to store charge.

$$Q = C \times V$$

- Current in a capacitor can be expressed as,

$$i = \frac{dQ}{dt} \Rightarrow i = C \frac{dV}{dt}$$

- **Energy stored in capacitor:**

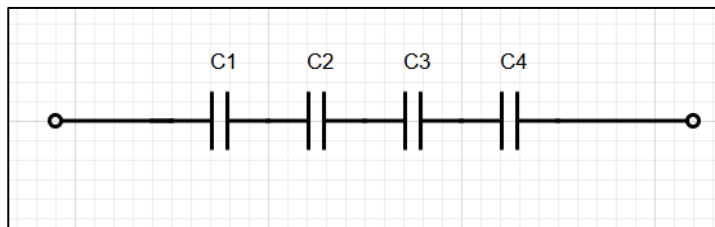
○ We know that $E = \int_0^t P dt$ and $P = V \times i$

$$\Rightarrow E = \int_0^t (V \times i) dt = \int_0^t \left(V \times C \frac{dV}{dt} \right) dt$$

$$\therefore E = \frac{1}{2} C [V^2(t) - V^2(0)]$$

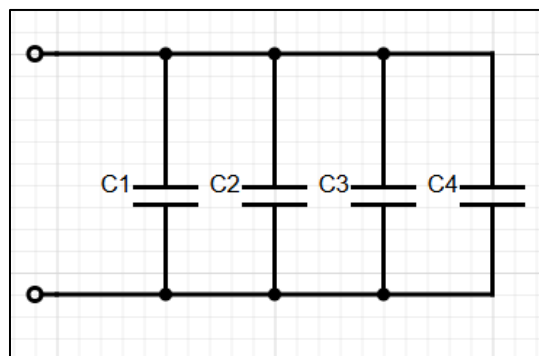
- **Series Connection:**

○ $C_{eq} = 1 / (1/C_1 + 1/C_2 + \dots + 1/C_n)$



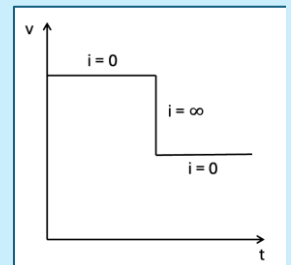
- **Parallel Connection:**

○ $C_{eq} = C_1 + C_2 + \dots + C_n$



NOTE

Voltage in a capacitor cannot change abruptly!



Current becomes infinite...

4) Inductor: -

- It is a device that stores energy in the form of moving charges (current).
- Inductance (L) is the measure of opposition to the change in current flowing caused by an external electromagnetic field.

$$V = L \frac{di}{dt}$$

- **Energy stored in inductor:**

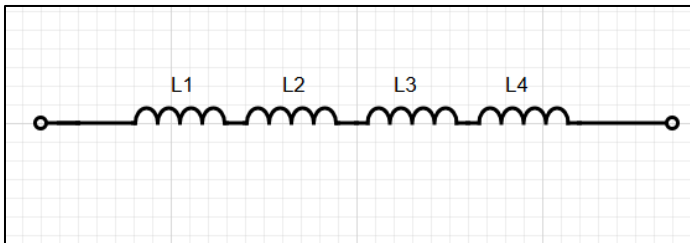
○ We know that $E = \int_0^t P dt$ and $P = V \times i$

$$\Rightarrow E = \int_0^t (V \times i) dt = \int_0^t \left(L \frac{di}{dt} \times i \right) dt$$

$$\therefore E = \frac{1}{2} L [i^2(t) - i^2(0)]$$

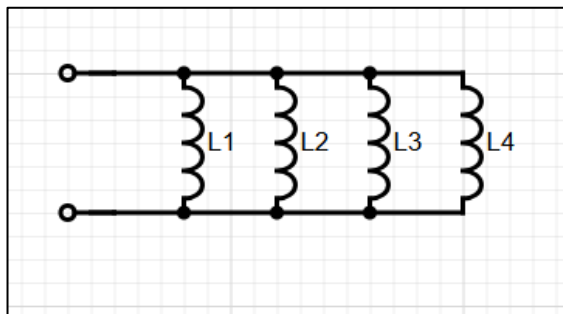
- **Series Connection:**

○ $L_{eq} = L_1 + L_2 + \dots + L_n$



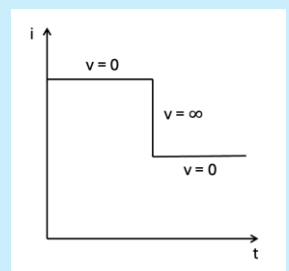
- **Parallel Connection:**

○ $L_{eq} = 1 / (1/L_1 + 1/L_2 + \dots + 1/L_n)$

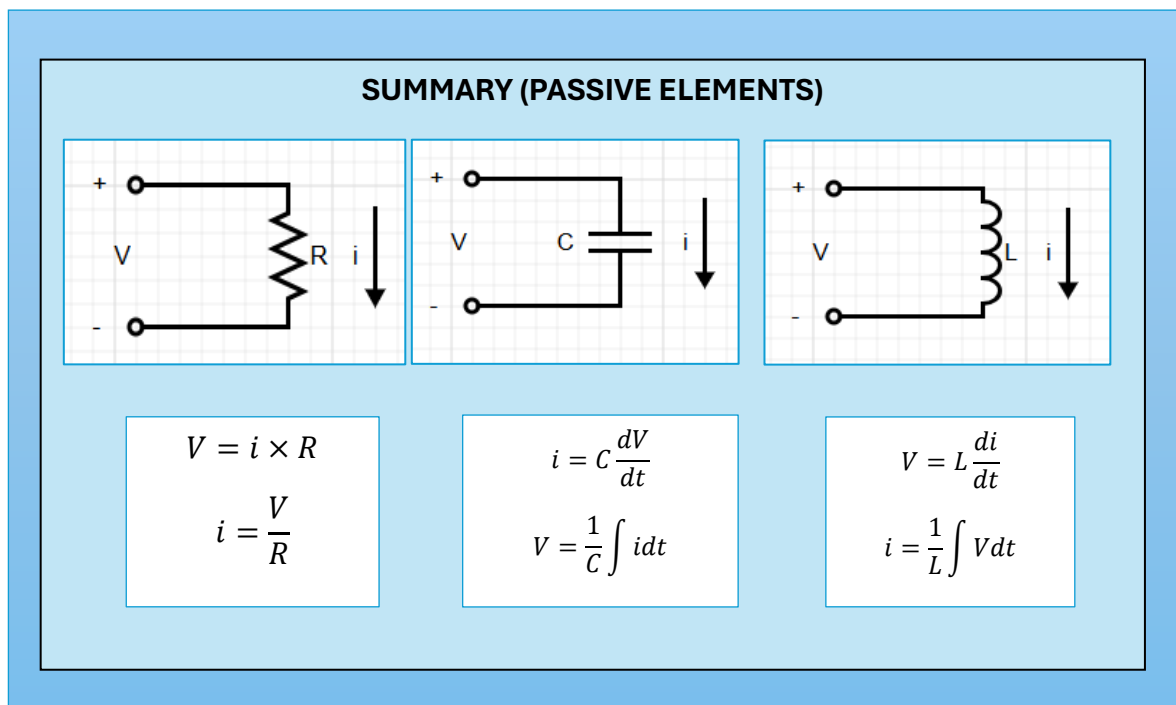


NOTE

Current in an inductor cannot change abruptly!



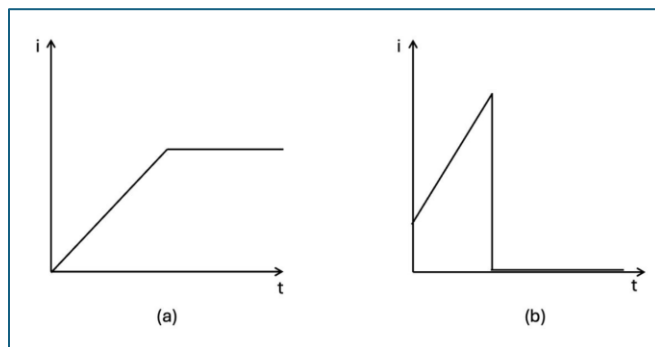
Voltage becomes infinite...



Related Examples

Example 1:

Which of the following can be the current through an inductor

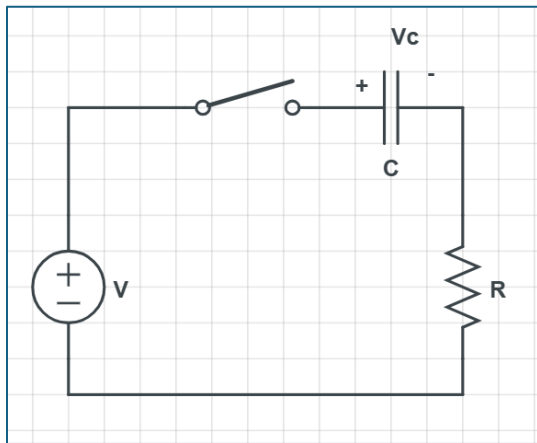


Solution:

Current shown in graph (a) can be the current through an inductor. The current changes abruptly in graph (b), this is not possible for an inductor.

Example 2:

If the switch closes at $t = 0$ and the capacitor is initially relaxed, find $i(0^+)$, $V_c(0^+)$, $dV_c/dt(0)$.



Solution:

Since $V_c(0^-) = 0 \rightarrow V_c(0^+) = 0$ (The switch just closes at $t=0$)

$$i(0^+) = \frac{V}{R}, \text{ Also } i(0^+) = C \frac{dV}{dt}(0^+)$$

$$\therefore \frac{dV}{dt}(0^+) = \frac{V}{RC}; i(0^+) = \frac{V}{R}; V_c(0^+) = 0$$

CIRCUIT ANALYSIS

Basic Circuit Examples

Example 1:

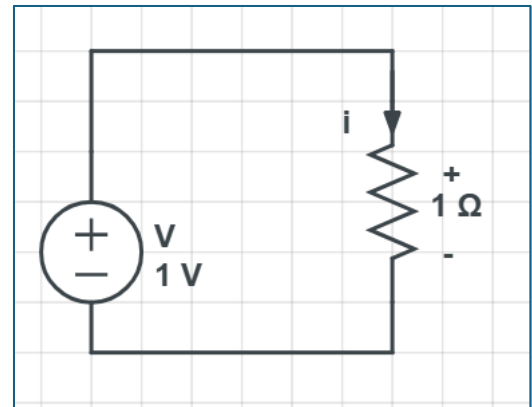
Find i for the circuit shown.

Solution:

Given that, $V = 1$ volt

The voltage source will provide a fixed voltage across the terminal, but current, i , will depend on the load connected which is the $1\ \Omega$ resistor,

$$\therefore i = \frac{V}{R} = \frac{1}{1} = 1\ A$$



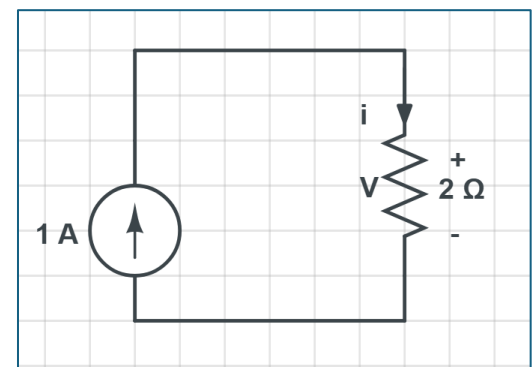
Example 2:

Find voltage across resistor (V) for the circuit shown.

Solution:

Given that current source provides $i=1A$, this current is fixed across the terminals, in this case the voltage depends on the load ($R = 2\ \Omega$),

$$\therefore V = i \times R = 1 \times 2 = 2\ Volts$$

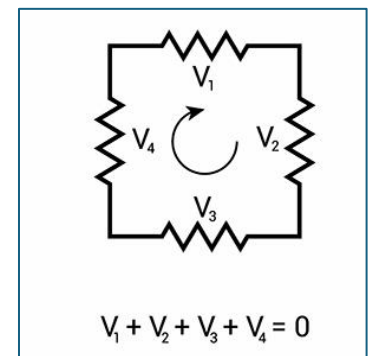


For solving and analyzing more complicated circuits we require certain theorems,

- KVL (Kirchoff's Voltage Law)
- KCL (Kirchoff's Current Law)

KVL (Kirchoff's Voltage Law)

- It states that the algebraic sum of voltages in a loop or closed path is zero.
- Sign Convention: When applying KVL, consider voltage drops (current flowing from higher to lower potential) as negative and voltage rises (sources) as positive.

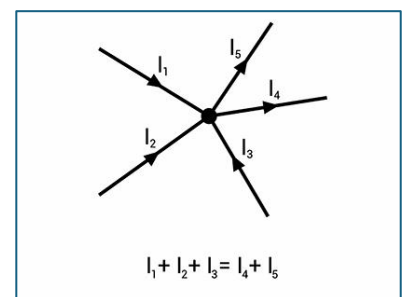


Mathematical Expression:

- $\sum V = 0$

KCL (Kirchoff's Current Law)

- KCL states that the algebraic sum of currents entering and leaving a node (or junction) in an electrical circuit is always equal to zero.
- It is based on the **law of conservation of electric charge**, which implies that charge



cannot accumulate at a node; all the current entering a node must leave the node.

Mathematical Expression:

$$\sum I_{in} = \sum I_{out} \quad \text{or} \quad \sum I = 0$$

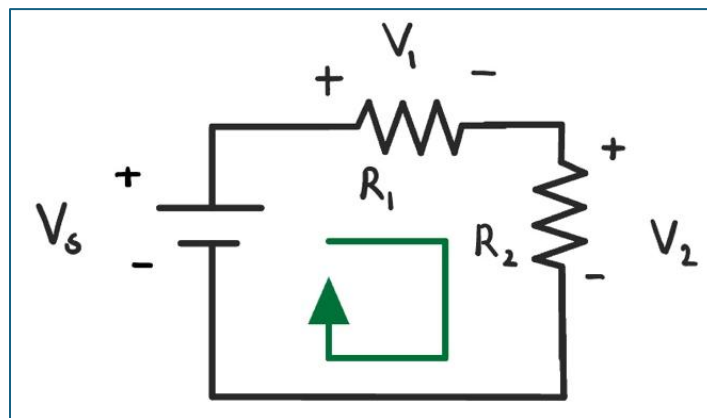
Where:

- I represents the electric currents at a node.
- Incoming currents are considered positive, and outgoing currents are considered negative (or vice versa based on convention).

Example Questions on Kirchhoff's Voltage Law (KVL):

Example 1:

Consider the electrical circuit shown below, which consists of multiple resistors and voltage sources arranged in a closed loop. Using Kirchhoff's Voltage Law (KVL), write the equation that represents the sum of voltages around the loop. (V_s is Voltage across the battery).



Solution:

- Ohms Law: $V = IR$
- KVL: $\sum V = 0$
- This is a series circuit, so the current remains the same throughout the entire circuit.

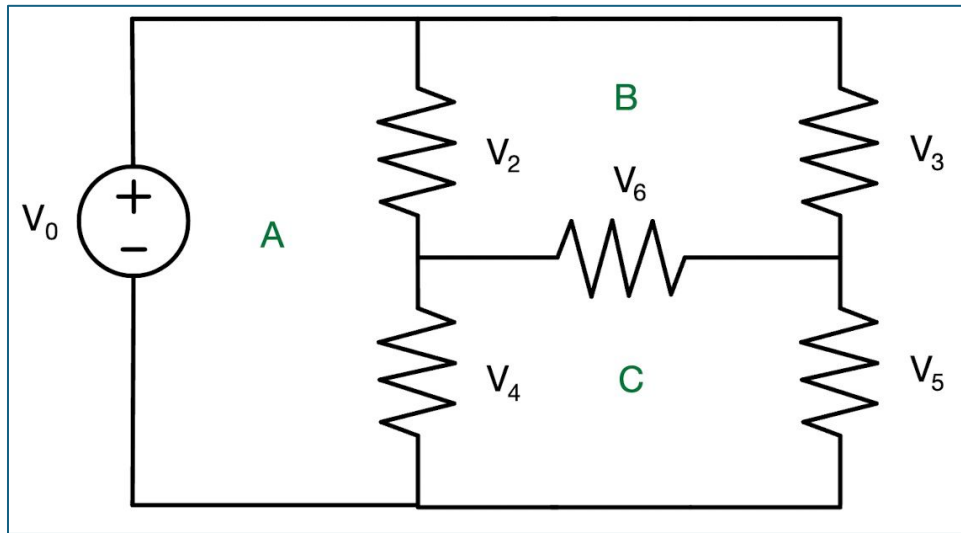
Applying KVL to the given circuit:

$$V_1 + V_2 - V_s = 0$$

$$IR_1 + IR_2 - V_s = 0$$

Example 2:

Consider the electrical circuit shown below, which consists of multiple resistors and voltage sources arranged in a closed loop. Using Kirchhoff's Voltage Law (KVL), write the equation that represents the sum of voltages around the linearly independent loops.



Solution:

Note: 7 loops are possible but not all of them are linearly independent.

The equations for the loops A, B, C, and D (A and B together) using KVL are:

$$\text{A: } V_2 + V_4 - V_0 = 0$$

$$\text{B: } V_3 - V_6 - V_2 = 0$$

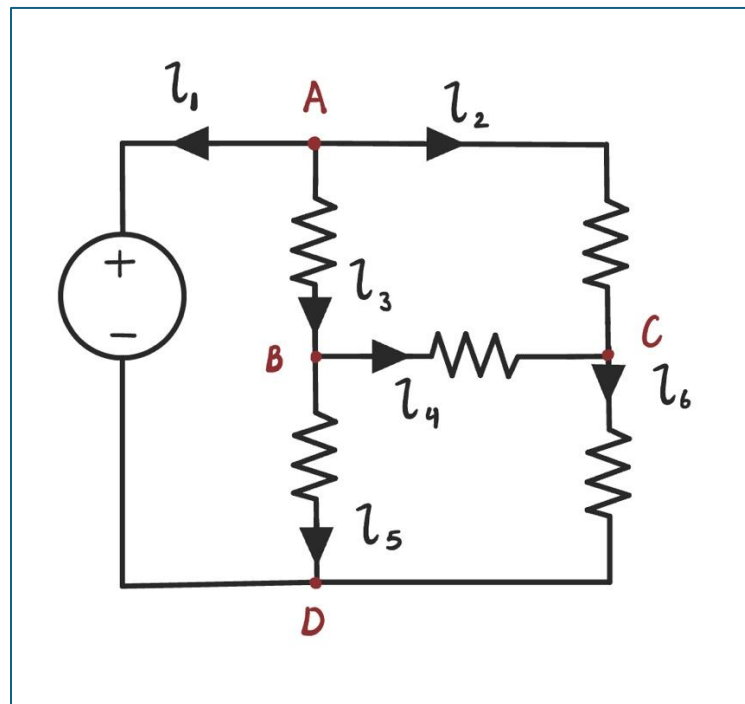
$$\text{C: } V_5 - V_4 + V_6 = 0$$

$$\text{D: } V_3 - V_6 + V_4 - V_0 = 0$$

Example Questions on Kirchhoff's Current Law (KCL):

Example 1:

Consider the following electrical circuit with **4 nodes**, where currents are flowing into and out of each node. Write the equation of each node.



Solution:

Step 1: Identify the Nodes and Currents (entering or leaving)

- Node A: Currents I_1 (leaving), I_2 (leaving), I_3 (leaving)
- Node B: Currents I_3 (entering), I_4 (leaving), I_5 (leaving)
- Node C: Current I_6 leaving, I_2 (entering), I_4 (leaving)
- Node D: Current I_5 (entering), I_6 (entering), I_1 (entering)

Step 2: Apply Kirchoff's Current Law (KCL) to Each Node

Node 1:

$$-I_1 - I_2 - I_3 = 0$$

Node 2:

$$I_3 - I_4 - I_5 = 0$$

Node 3:

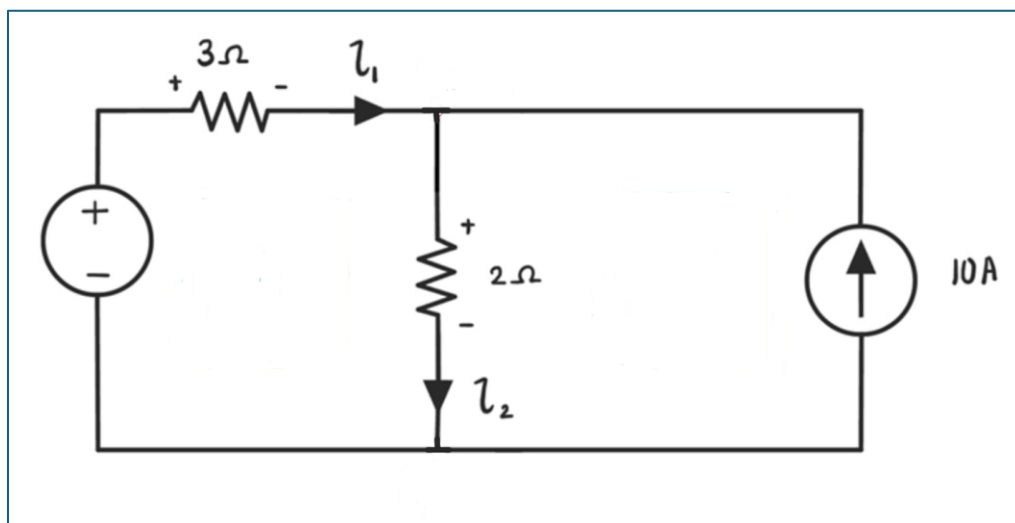
$$-I_6 + I_2 + I_4 = 0$$

Node 4:

$$-I_5 - I_6 + I_1 = 0$$

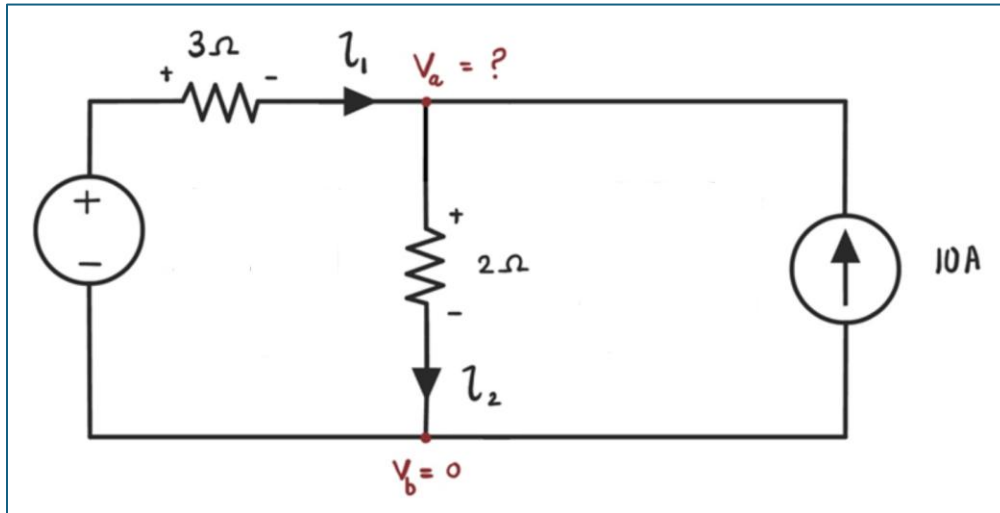
Example 2:

A circuit is shown below. Find I_1 . (DC source provides 15 volts)



Solution:

Using Node Method:



Step 1: Identify the Nodes

- Node A where we want to find V_a
- Node B which is the reference (ground) node ($V_b = 0$)

Step 2: Apply KCL at Node A, and Solve for V_a

Current through each branch can be written using Ohms law:

$$I = \frac{V_a - V_{neighbor}}{R}$$

Apply KCL at node A:

$$\frac{V_a - 0}{2} - 10 + \frac{V_a - 15}{3} = 0$$

Solving for V_a we get:

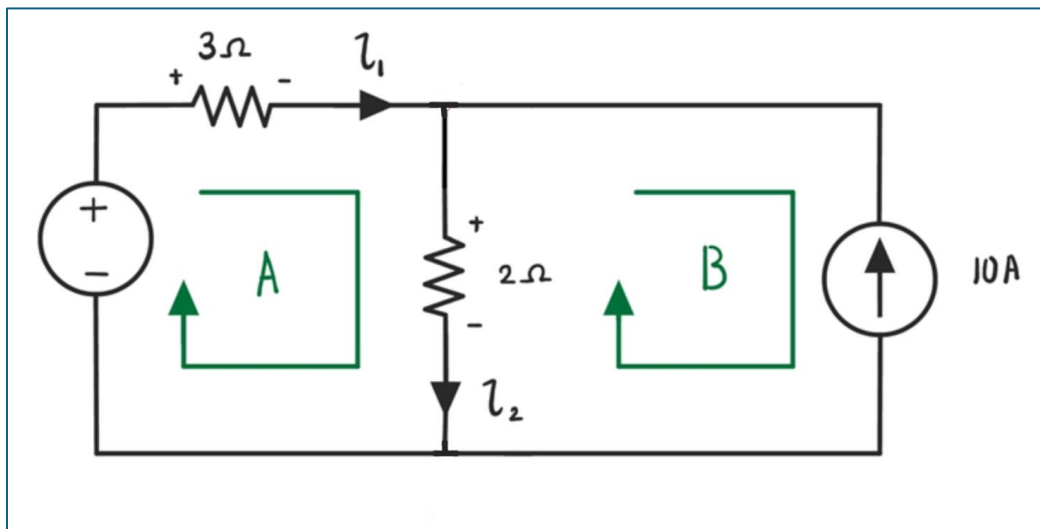
$$V_a = 18 \text{ volts}$$

Step 3: Find I_1

From ohms law we get:

$$I_1 = \frac{15 - V_a}{3} = \frac{15 - 18}{3} = -1 \text{ A}$$

Another Method to Solve Example 2 is the **Loop (Mesh) Method**:



Step 1: Define Loop Currents

- Loop 1: 15V voltage source, resistors of 3Ω , and 2Ω

Step 2: Apply Kirchhoff's Voltage Law (KVL): $\sum V = 0$

$$3I_1 + (2)(I_1 - I_2) - 15 = 0$$

Solving for I_1 and I_2 we get:

$$I_1 = -1 \text{ A}$$

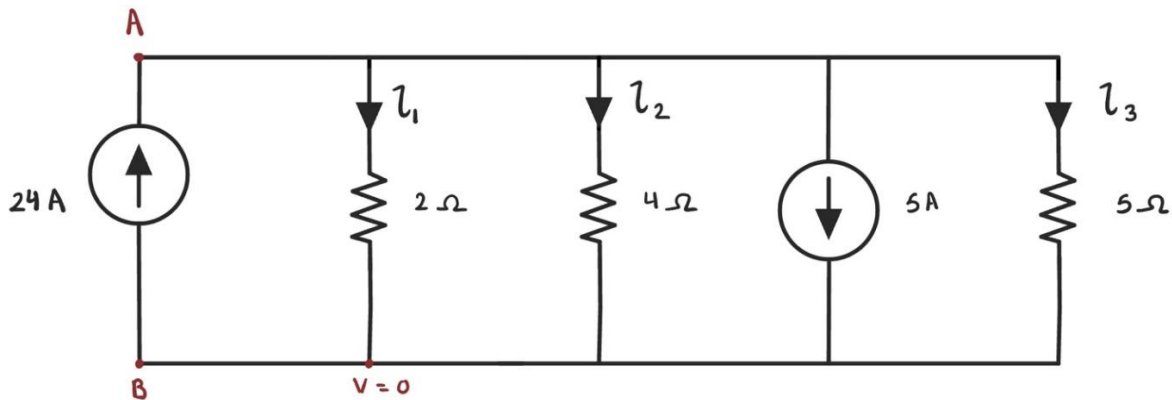
$$I_2 = -10 \text{ A}$$

NOTE

- Node method uses KCL
- Mesh analysis/Loop method uses KVL

Example 3:

Consider the following electrical circuit. Find V_{AB} , I_1 , I_2 , I_3



Step 1: Apply Kirchhoff's Current Law (KCL) at Node A, and Solve for V_A

Note:
$$I = \frac{V_a - V_{neighbor}}{R}$$

$$\sum I_{in} = \sum I_{out}$$

$$\frac{V_A - 0}{2} + \frac{V_A - 0}{4} + 5 + \frac{V_A - 0}{5} = 24$$

$$V_A = 20 \text{ volts}$$

Step 2: Find I_1 , I_2 , I_3

- From Ohms Law we know that: $I = \frac{V}{R}$
 - V: voltage across each resistor (V_A is this case)
 - R: Resistance

Finding I_1 , I_2 , I_3 :

$$I_1 = \frac{20}{2} = 10A$$

$$I_2 = \frac{20}{4} = 5A$$

$$I_3 = \frac{20}{5} = 4A$$
